

CrossWeaver: Towards Efficient Cross-Modal Interweaving and Decoupling for Weakly-Aligned Multispectral Object Detection

Haitian Yang^{1,2*} Juan Fang¹ Yiren Zhu² Xudong Zhao³ Yufei Guo⁴

Xiaohan Zhang⁵ Xiaoxing Hu³ Xue Yang⁶ Qi Ming^{1†}

¹Beijing University of Technology ²ShanghaiTech University ³Beijing Institute of Technology

⁴Intelligent Science and Technology Academy of CASIC

⁵Zhejiang University ⁶Shanghai Jiao Tong University

Abstract

While single-modal object detection has made significant progress, real-world perception increasingly depends on multimodal data to capture richer visual cues. However, multimodal object detection under weak spatial alignment remains challenging. In this work, we revisit the problem from two key perspectives: (1) quantization error caused by misaligned multimodal fusion, and (2) balancing modality similarity and complementarity for robust feature integration. Building upon these insights, we propose a novel Cross-Modal Semantic Weaving Network (CrossWeaver) for weakly aligned multimodal object detection. Specifically, we design the Modality-Contrastive Shared Encoder (MCSE) to extract shared semantic representations across modalities under contrastive supervision. Then, we propose the Modality-Conditioned Deformation Adapter (MCDA) to dynamically modulate spatial sampling fields for adaptive geometric transformations and inject lightweight modality-specific priors. Finally, the Cross-Hierarchical Synergistic Network (CHSN) establishes bidirectional interactions between shared semantic and modality-aware features to mitigate quantization errors induced by weak alignment. Extensive experiments show that CrossWeaver achieves state-of-the-art performance on weakly aligned multimodal object detection benchmarks. Our method achieves the mAP_{50} of 86.8% on DroneVehicle dataset. Our code: <https://github.com/zhoushang11/CrossWeaver>.

1. Introduction

Computer vision has become an essential part of many real-world applications, including autonomous driving [47], document analysis [26, 27], industrial inspection [12], medical image segmentation [21], and remote sensing [22, 23].

*Work done during internship at Beijing University of Technology.

†Corresponding author: Qi Ming, mingqi@bjut.edu.cn

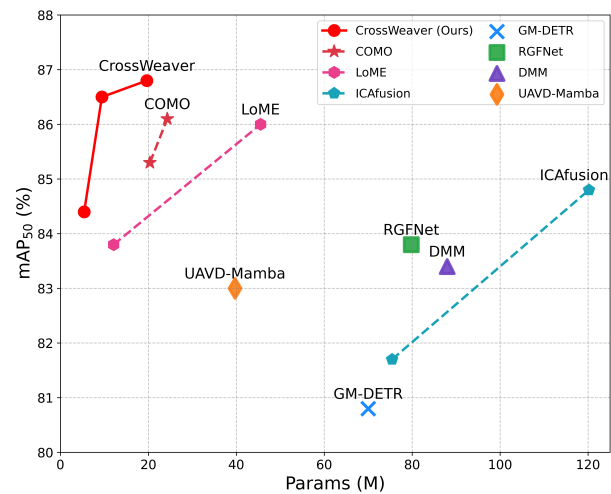


Figure 1. Comparison with state-of-the-art methods on the DroneVehicle dataset in terms of mAP_{50} and model size.

Among its core tasks, object detection plays a fundamental role by connecting high-level scene understanding with low-level perception. With the success of convolutional networks [29, 30] and vision transformers [5, 20], single-modal detectors have achieved impressive accuracy and efficiency. However, in complex environments with varying illumination, occlusion, or cluttered backgrounds, using only one sensing modality limits the robustness and generalization.

To address these limitations, the multimodal object detection (MMOD) task has garnered increasing attention. By integrating complementary visual cues from heterogeneous sensors such as visible, infrared, depth, and radar, multi-modal detection seeks to enhance scene understanding under adverse conditions. Among these cases, infrared-visible image object detection serves as a representative application scenario [6, 14, 40]. Early research [37] focused on straightforward feature concatenation or late fusion strategies, which often struggled with modality imbal-

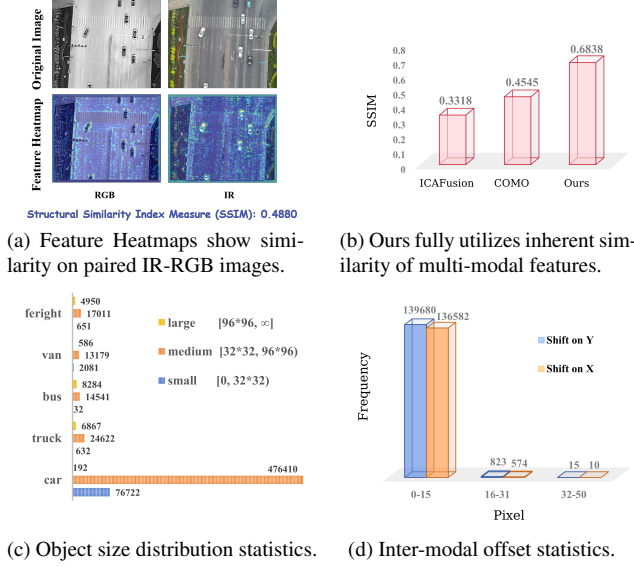


Figure 2. Analysis and visualization results of objects in the DroneVehicle dataset. (a) The dual backbone produces feature maps with notable inherent similarity. (b) A shared backbone is sufficient to effectively exploit cross-modal homogeneous features. (c) The dataset is dominated by small and medium-sized objects. (d) 99.98% of inter-modal spatial offsets are less than 32 pixels.

ance and weak alignment issues. Subsequent studies explored deep cross-modal fusion and adaptive weighting to dynamically integrate features, yet most still assume perfect pixel-level alignment between modalities [35]. Recently, most work has investigated alignment-aware fusion or contrastive representation learning to solve the weak alignment problem in MMOD task [1, 2, 44].

Despite recent progress in MMOD, there are still significant challenges. First, we observe the **inefficient balancing of modality similarity and complementarity** lies in mainstream frameworks, which employ dual feature extractors to encode infrared and visible image features separately leading to a substantial increase in model sizes [1, 4]. This design implicitly emphasizes capturing heterogeneous features, while overlooking their dominant similarity. As shown in Fig. 2a and 2b, we find that feature activation regions from dual-backbone ICAFusion [32] exhibit high similarity, achieving a SSIM of 0.4880. It reveals that the features are inherently homogeneous. Therefore, a parameter-shared backbone is sufficient to effectively extract these homogeneous representations. Subsequently, we posit that lightweight, complementary features should be injected not just as a supplement, but as a crucial driver for precise spatial refinement.

Second, we note the **quantization errors** in multi-scale predictions for weakly aligned cross-modal detection. As illustrated in Fig. 2c, the DroneVehicle [34] dataset is dom-

inated by small, weakly aligned objects. These objects are particularly vulnerable to spatial information loss after feature downsampling. For instance, when an object is smaller than 32×32 pixels, its spatial representation and the corresponding inter-modal offset are often quantized into a single feature point, rendering the precise spatial information unrecoverable from high-level features alone. Therefore, we suggest that low-level features should guide high-level features for spatial refinement. Meanwhile, Fig. 2d shows that the majority of spatial offsets are minor. This allows high-level features to be naturally insensitive to offset noise, enabling them to serve as pure and robust cross-modal semantic guidance that is essential for accurate classification.

Based on the above observations, we propose a Cross-Modal Semantic Weaving Network (CrossWeaver) to address the challenges of parameter redundancy and quantization error. First, we employ a parameter-shared Modality-Contrastive Shared Encoder (MCSE) to extract preliminary cross-modal homogeneous feature representations, supervised by contrastive learning to ensure effective semantic alignment. Next, we design a Modality-Conditioned Deformation Adapter (MCDA), which dynamically modulates heterogeneous features by embedding lightweight and modality-sensitive information to generate precise spatial guidance that complements the shared representations and drives spatial refinement. Finally, we construct a Cross-Hierarchical Synergistic Network (CHSN) to enable bidirectional semantic guidance between shared and modality-specific features. Specifically, the top-down path propagates semantic guidance to improve detection accuracy, while the bottom-up path injects spatial details to perform spatial refinement and compensate for quantization loss.

As shown in Fig. 1, our method achieves mAP_{50} of 86.8% on the DroneVehicle dataset, demonstrating a favorable trade-off between accuracy and efficiency, outperforming numerous existing approaches. Furthermore, we validate its deployment capability on the NVIDIA Jetson NX platform, where our lightweight variant achieves real-time processing speed up to 56.39 FPS. This confirms our method as a practical solution for remote sensing applications. Our main contributions are summarized as follows:

- We reveal the dominance of shared representations in IR-RGB object detection and design a MCSE with contrastive supervision for efficient cross-modal homogeneous feature extraction.
- To emphasize modality heterogeneity for modality-specific detection, we propose the MCDA, which incorporates lightweight priors to construct modality-sensitive spatial offset fields capturing weakly aligned semantics.
- We analyze quantization errors on small and misaligned objects and propose a CHSN that enhances accuracy and localization through bidirectional semantic guidance and spatial refinement.

2. Related Work

2.1. Generic and Multi-Modal Object Detection

The foundation of modern object detection rests on convolutional neural networks, which evolved along two primary trajectories: the two-stage paradigm pioneered by Faster R-CNN [30] and the one-stage approach introduced by YOLO [29]. However, all these single-modal approaches share a fundamental limitation: they are constrained by the inherent vulnerabilities of a single sensor. For example, the RGB cameras inherently limit robustness in adverse conditions. This critical vulnerability spurred the development of MMOD, typically fusing visible and infrared data [9] to improve robustness. Early attempts at simple image concatenation or late-stage result merging [24] were quickly superseded by feature-level fusion, a paradigm pioneered by Halfway Fusion [15], which demonstrated superior results by integrating features mid-network. This fusion-centric development has continued to evolve, adopting sophisticated architectures like the Transformer, introduced by CFT [28], and more recently exploring efficient alternatives like State Space Models [11, 13, 49]. However, this entire MMOD task has largely progressed under the critical and often unrealistic assumption that the input modalities are perfectly aligned.

2.2. Weakly-Aligned Multi-Modal Detection

In real-world applications, sensors are rarely perfectly aligned, creating a weak alignment problem that can break standard fusion methods [1, 38]. Research to solve this has generally followed two paths. The first is explicit alignment, where the network tries to physically warp one image to match the other, either through traditional image registration [50] or by learning a deep learning-based deformation field [31]. The second and more recent path is adaptive alignment. Here, instead of warping the image, the network learns to adapt its features during the fusion process. A pioneering data-driven approach by AR-CNN [41] used a module to predict proposal region offsets. A key enabling technology for this is the Deformable Convolution (DCN) [3], which allows the network to learn where to sample features from, effectively compensating for misalignment on the fly. Building on this, methods like OAFA [1] use misalignment-invariant features to guide this deformable sampling, while others use Vision Transformer [32], wavelet analysis [38] or Mamba [49] to handle discrepancies and misalignment. Despite this progress, developing solutions that are simultaneously robust to misalignment, computationally efficient, and architecturally simple remains challenging.

3. Methodology

3.1. Modality-Contrastive Shared Encoder

Current approaches in MMOD often employ dual encoders for the infrared and visible modalities. As illustrated in Fig. 2a, our key insight is that features extracted from both modalities are already homogeneous, which strongly suggests that a heavy dual-backbone architecture is redundant. Therefore, we propose a parameter-shared encoder $\mathcal{E}_{shared}$ to efficiently capture these homogeneous features F_{homo} in a unified representation.

$$F_{homo}^{RGB}, F_{homo}^{IR} = \mathcal{E}_{shared}(I_{RGB}, I_{IR}) \quad (1)$$

where I_{IR} and I_{RGB} denote a pair of weakly-aligned infrared and visible images. Although the features are homogeneous from both modalities, the raw visible and thermal data possess distinct statistical properties. To account for this distinction, we maintain separate batch normalization (BN) layers for each modality to preserve modality-specific statistics, rather than naively sharing all parameters.

$$\begin{aligned} F_{homo}^{RGB} &= \text{BN}_{RGB}(\text{Conv}_{shared}(I_{RGB})) \\ F_{homo}^{IR} &= \text{BN}_{IR}(\text{Conv}_{shared}(I_{IR})) \end{aligned} \quad (2)$$

Our observation in Fig. 2d is that the vast majority of inter-modal offsets are less than 32 pixels, which renders the high-level P_5 feature map inherently insensitive to these misalignments due to its large stride. To ensure this P_5 representation serves as robust semantic guidance, we enforce true semantic alignment for the subsequent top-down semantic guidance.

Therefore, we employ contrastive learning for this purpose. Specifically, we introduce an InfoNCE loss $\mathcal{L}_{contrast}$ [25], which effectively pulls corresponding IR–RGB features closer together in the embedding space while pushing apart all non-corresponding pairs.

$$\mathcal{L}_{contrast} = -\log \frac{\exp(\text{sim}(F_{homo}^{RGB,5}, F_{homo}^{IR,5})/\tau)}{\sum_k \exp(\text{sim}(F_{homo}^{RGB,5}, F_k^{IR,5})/\tau)} \quad (3)$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity, τ is a temperature parameter, and $F_k^{IR,5}$ represents the negative samples from other images within the batch.

3.2. Modality-Conditioned Deformation Adapter

While the MCSE backbone provides a strong, homogeneous F_{homo} representation, it is insufficient for precise localization due to quantization-induced information loss.

To compensate for this spatial information loss, we also extract the unique, heterogeneous features F_{het} , which will serve as the low-level spatial guidance to refine the high-level semantic features in the bottom-up path. We propose

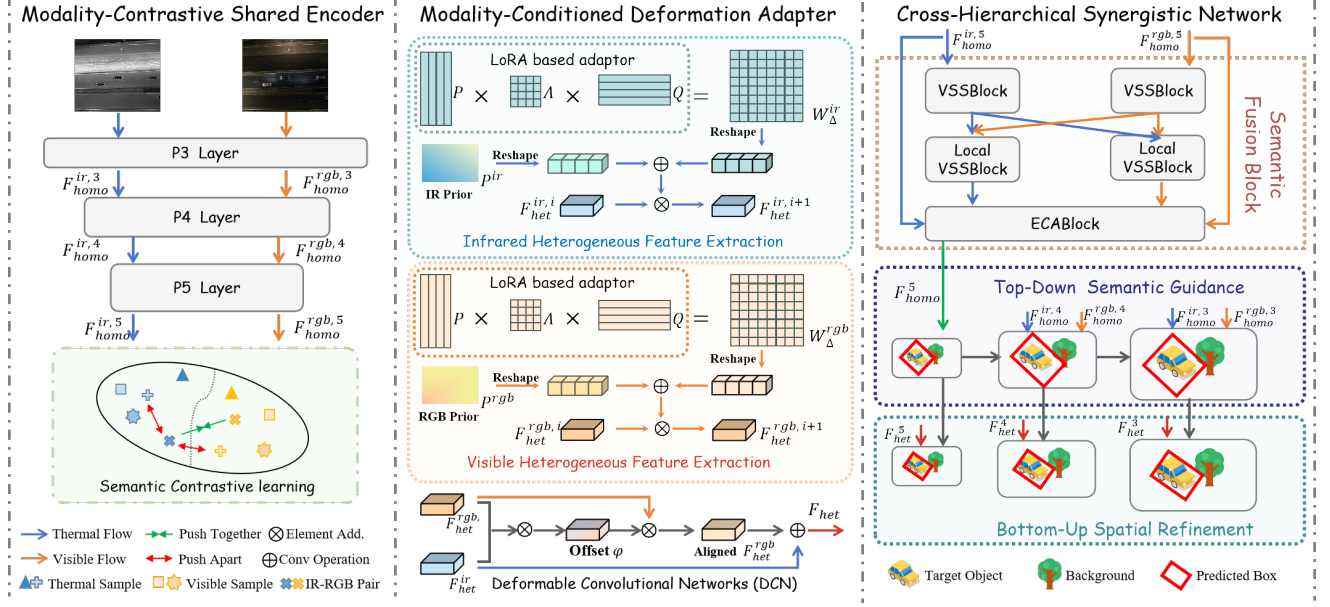


Figure 3. Overall architecture of our proposed CrossWeaver. The MCSE first extracts efficient, homogeneous features F_{homo} and establishes robust semantic guidance at P_5 using contrastive learning. In parallel, a lightweight MCDA extracts and aligns the heterogeneous features F_{het} that capture spatial details lost to quantization. The CHSN employs a top-down semantic guidance path to improve detection accuracy, and a bottom-up spatial refinement path to perform spatial refinement and compensate for quantization loss.

the MCDA as the lightweight extractor dedicated to this task. The MCDA is composed of two components: a Low-Rank Adaptation (LoRA) [7] based adaptor to efficiently extract sparse feature modulations, and a modality-specific prior module providing unique cues for each modality. These components are then added to extract the final F_{het} representation, which is ready for bottom-up spatial refinement path in the next step.

We now detail the MCDA components. First, the LoRA-based adaptor parameterizes its extraction weights W_{Δ} using low-rank decomposition rather than relying on a full, dense weight matrix.

$$W_{\Delta} = B \odot (A \cdot E) \quad (4)$$

where $A \in \mathbb{R}^{r \times (c_{in} \cdot k^2)}$ and $B \in \mathbb{R}^{c_{out} \times r}$ are the low-rank matrices, and $E \in \mathbb{R}^{r \times 1}$ is the learnable importance vector. The rank r is a small hyperparameter, ensuring that we capture these modality-specific modulations with minimal parameter overhead.

In parallel to the LoRA-based adaptor, the second component of the MCDA is modality-specific prior module, which provides unique, prior knowledge P^m , $m \in \{RGB, IR\}$. For RGB images, we extract texture information using the Laplacian operator ∇^2 , a filter that highlights regions of rapid intensity change.

$$P^{RGB} = |\nabla^2(I_{RGB})| \quad (5)$$

For IR images, we capture thermal characteristics by combining gradient information P_{grad}^{IR} with thermal continuity P_{cont}^{IR} . This gradient information is derived using the Sobel operator $\mathcal{S}$ for edge detection.

$$\begin{aligned} P_{grad}^{IR} &= |\mathcal{S}_x(I_{IR})| + |\mathcal{S}_y(I_{IR})| \\ P_{cont}^{IR} &= \exp\left(-\frac{\text{Var}_{local}(I_{IR})}{\sigma^2}\right) \\ P^{IR} &= [P_{grad}^{IR}, P_{cont}^{IR}] \end{aligned} \quad (6)$$

where Var_{local} computes local variance within a 5×5 window, and $\sigma = 0.05$ controls the sensitivity. Finally, these priors P^m are projected by a learnable 1×1 convolution $\text{Conv}_{1 \times 1}$ to match the feature dimensions at multiscale feature levels, yielding the final prior features $P^{m,i}$.

We combine LoRA-based adaptor W_{Δ} and the Modality-Specific Prior $P^{m,i}$ to extract the heterogeneous features F_{het} for each modality.

$$\begin{aligned} F_{het}^{RGB,i} &= \text{Conv}(I_{RGB}; W_{\Delta}^{RGB} + \alpha \cdot P^{RGB,i}) \\ F_{het}^{IR,i} &= \text{Conv}(I_{IR}; W_{\Delta}^{IR} + \alpha \cdot P^{IR,i}) \end{aligned} \quad (7)$$

where the scaling factor α sets to 1.0. The extracted heterogeneous features $F_{het}^{RGB,i}$ and $F_{het}^{IR,i}$ are still weakly-aligned. To enable F_{het} to provide more precise low-level guidance, we employ Deformable Convolution Networks (DCN) [3] to explicitly align them. As shown in Fig. 3, the DCN first learns these offsets from both modality, then

warps $F_{het}^{RGB,i}$ to match $F_{het}^{IR,i}$, and finally adds them to produce an aligned heterogeneous feature map F_{het}^i .

$$F_{het}^i = \text{DCN}(F_{het}^{IR,i}, F_{het}^{RGB,i}) \quad (8)$$

These aligned heterogeneous features F_{het}^i serve as a crucial supplement to the shared encoder by capturing the fine-grained, modality-specific details lost to quantization, which provide the essential low-level guidance for the subsequent bottom-up refinement path.

3.3. Cross-Hierarchical Synergistic Network

We propose the CHSN to execute this dual-guidance strategy: (1) a top-down path provides semantic guidance from P_5 level to enhance detection accuracy, and (2) a bottom-up spatial refinement path injects the lost spatial details F_{het} to correct fine-grained representations.

Top-Down Semantic Guidance. This top-down path begins at the P_5 level. Based on our analysis in Fig. 2d, the vast majority of inter-modal offsets are smaller than 32 pixels. This means that P_5 level is offset-insensitive and retains semantic information. Therefore, we design the semantic fusion block $\mathcal{G}_{sem}$ to deeply fuse the homogeneous features $F_{homo}^{RGB,5}$ and $F_{homo}^{IR,5}$ into robust semantic guidance shown in Fig. 3.

$$F_{fused}^5 = \mathcal{G}_{sem}(F_{homo}^{RGB,5}, F_{homo}^{IR,5}) \quad (9)$$

The semantic fusion block first refines each modality features using Visual State Space (VSS) blocks [18] to model global context. Then, a LocalVSS module [8] performs efficient cross-modal interaction. This yields F_{fused}^5 , a semantically robust and globally aligned feature map that serves as the starting point for the top-down path.

This F_{fused}^5 feature now propagates its strong semantic guidance downwards. At each subsequent level, this high-level prior fuses with the local, scale-specific homogeneous features $F_{homo}^{RGB,i}$ and $F_{homo}^{IR,i}$. Since these local features are still weakly-aligned, we employ DCN to align and fuse them. This is then fused with the upsampled semantic guidance using a C2f block from yolov8 [10].

$$F_{guide}^i = \text{C2f}[\text{Up}(F_{guide}^{i+1}), \text{DCN}(F_{homo}^{IR,i}, F_{homo}^{RGB,i})] \quad (10)$$

where $i \in \{3, 4\}$. $\text{Up}(\cdot)$ denotes a $2 \times$ upsampling operation. F_{guide}^5 is initialized as F_{fused}^5 . This cascade ensures semantic information are propagated while spatial alignment is progressively refined at each scale.

Bottom-Up Spatial Refinement. The bottom-up refinement path is designed to solve the quantization problem. It injects the precise, heterogeneous features F_{het} from MCDA at each scale, correcting the spatial inaccuracies lost to quantization.

This refinement process begins at P_3 and propagates upward. The initial refined feature F_{refine}^3 is computed by

injecting the spatial details F_{het}^3 directly into the semantic guidance F_{guide}^3 . For the subsequent levels $j \in \{4, 5\}$, the final refined feature F_{refine}^j is computed by fusing the downsampled path with the combined guidance features.

$$F_{refine}^j = \text{C2f}[\text{Down}(F_{refine}^{j-1}), F_{guide}^j + F_{het}^j] \quad (11)$$

where F_{guide}^j is the output from the top-down path, F_{refine}^j is the final refined feature map, and $\text{Down}(\cdot)$ denotes a $2 \times$ downsampling operation.

For small objects, this path carries the fine-grained spatial information that is nullified at the P_5 level back up to the high-level detection heads. For large objects, this path provides the modality-specific boundary details by F_{het} necessary to refine localization. This bottom-up path thereby effectively complements the existing top-down guidance, ensuring the features are both semantically rich and spatially accurate for the detection head.

3.4. Loss Functions

Our network learns two distinct tasks: the primary goal of detection and the auxiliary task of enforcing semantic alignment in our shared encoder. Therefore, we combine a standard detection loss $\mathcal{L}_{det}$ with our semantic alignment loss $\mathcal{L}_{InfoNCE}$ using a balance hyperparameter $\lambda_{contrast}$.

$$\mathcal{L}_{total} = \mathcal{L}_{det} + \lambda_{contrast} \cdot \mathcal{L}_{InfoNCE} \quad (12)$$

The detection loss $\mathcal{L}_{det}$ follows the standard set by [10], which composes a weighted sum of three components: the CIoU loss for bounding box regression $\mathcal{L}_{box}$, the binary cross-entropy for classification $\mathcal{L}_{cls}$, and the distribution focal loss for box refinement $\mathcal{L}_{dfl}$.

$$\mathcal{L}_{det} = \mathcal{L}_{box} + \mathcal{L}_{cls} + \mathcal{L}_{dfl} \quad (13)$$

We empirically set the balancing hyperparameter $\lambda_{contrast}$ to a value of 0.1 to ensure both strong detection performance and robust semantic feature alignment.

4. Experiments

4.1. Dataset

We evaluate our method on three benchmarks with varying alignment conditions.

DroneVehicle. [34] Our primary benchmark is the DroneVehicle dataset, a large-scale aerial dataset of 28,439 pairs featuring weakly-aligned images under diverse lighting conditions. Its high prevalence of small and medium objects makes it the ideal testbed for validating our approach against the core challenges of weak alignment.

M3FD. [16] To test generality and robustness, we employ the M3FD dataset, which provides well-aligned vehicle-mounted data to verify performance under ideal conditions.

Table 1. Component ablation study on the DroneVehicle dataset to evaluate our proposed CrossWeaver model. It consists of three main modules: MCSE, MCDA and CHSN. The best results are indicated in **bold**.

MCSE		MCDA		CHSN			mAP ₅₀ (%)				mAP (%)	Params (M)	GFLOPs
Shared	ConL	Adaptor	Prior	Fusion	DCN	Inject	all	small	medium	large			
							84.1	16.2	80.6	47.1	62.6	16.50	42.2
✓		✓	✓	✓	✓	✓	86.5	18.5	81.9	48.8	65.1	19.67	65.7
✓	✓	✓	✓		✓	✓	85.6	17.3	82.0	46.7	63.6	12.34	20.2
✓	✓			✓	✓		85.3	17.2	81.7	47.0	64.0	18.96	64.0
✓	✓	✓	✓	✓		✓	86.2	17.8	82.1	48.0	65.2	19.50	64.6
✓	✓	✓		✓	✓	✓	86.0	17.6	81.5	47.6	64.8	19.66	65.7
✓	✓	✓	✓	✓	✓	✓	86.8	18.6	82.5	48.8	65.5	19.67	65.7

Shared signifies the switch from the dual-backbone to our shared-backbone architecture. **Fusion** denotes the semantic fusion block. **ConL** applies a contrastive learning on MCSE. **Adaptor** is a lightweight LoRA-based feature extractor, and **Prior** represents the modality-specific prior information. **DCN** is a Deformable Convolutional Network for offset prediction, while **Inject** is to feed heterogeneous information back for high-level refinement.

The M3FD dataset is sampled from video, resulting in high intra-scene similarity between adjacent frames. Random splits can thus suffer from information leakage and lead to overestimated generalization. We therefore follow the splits provided by [42] to ensure a valid evaluation.

DVTOD. [33] The DVTOD dataset presents a more demanding misalignment scenario and functions as an essential benchmark to test the limits of our alignment mechanisms, specifically evaluating their capacity to manage large-scale, non-rigid spatial discrepancies.

4.2. Experimental Setup

Implementation Details. Our CrossWeaver is trained from scratch on NVIDIA 4090 GPUs. For the large-scale DroneVehicle dataset, we train for 150 epochs using an SGD optimizer with a cosine annealing schedule, an 800×800 input resolution, and a 5-epoch warmup, disabling mosaic augmentation for the final 10 epochs. For the smaller M3FD and DVTOD datasets, we train for 300 epochs using an AdamW optimizer.

Evaluation Metrics. We use mean Average Precision (mAP) for evaluation, though the specific reported metrics vary by dataset protocol. On our primary DroneVehicle benchmark, we report mAP₅₀, mAP, total parameters (Params), and GFLOPs to evaluate both accuracy and efficiency. We also provide a detailed mAP analysis for small, medium, and large objects to validate our design. For M3FD, we report mAP₅₀ and mAP, while for DVTOD, we report mAP₅₀ consistent with its standard protocol.

4.3. Ablation Studies

Component Ablation. We conduct a comprehensive component ablation study to validate our design choices, with detailed results presented in Table 1. Our full CrossWeaver model achieves 86.8% mAP₅₀, which is a significant 2.7% improvement over the 84.1% mAP₅₀ from the standard dual-backbone baseline. This performance gain is directly

attributable to our core architectural innovations addressing parameter redundancy and quantization error. The most critical component is our proposed MCDA. As shown in the fourth row, disabling the MCDA module causes a substantial performance drop to 85.3% mAP₅₀. The 1.5% decrease, particularly the corresponding drop in small-object mAP₅₀ from 18.6% to 17.2%, explicitly validates our core hypothesis that the MCDA is essential for re-injecting the heterogeneous spatial details lost to quantization. Furthermore, the importance of our high-level semantic guidance is also validated. As demonstrated in the third row, removing the semantic fusion block causes a significant drop in mAP₅₀ from 86.8% down to 85.6%, which confirms that a robust semantic starting point is critical for the top-down guidance path.

Analysis of MCDA Rank Strategy. We investigate the optimal rank selection for the MCDA adaptor in Table 3. Our proposed dynamic rank strategy, transitioning from $r = 9$ to 6, achieves the optimal 86.8% mAP₅₀ and outperforms all fixed-rank configurations, including the higher-capacity $r = 12$ strategy. The dynamic strategy allows the network to capture a wide range of sparse modulations at earlier stages and subsequently refine them into a more compact, heterogeneous representation.

4.4. Comparison with State-of-the-Art Methods

We provide a comprehensive comparison between our CrossWeaver family and existing state-of-the-art methods on the DroneVehicle dataset, with results detailed in Table 2. Our largest model, CrossWeaver-S, achieves a new state-of-the-art performance of 86.8% mAP₅₀ and 65.5% mAP. This result is particularly noteworthy as it surpasses strong, pretrained, dual-backbone competitors like COMO [13], while our model is trained entirely from scratch. Our approach successfully mitigates the quantization error and also balances modality similarity and complementarity for robust feature integration. Furthermore,

Table 2. Comparison with state-of-the-art methods on the DroneVehicle dataset. The best and second-best results are indicated in **bold** and underlined, respectively. ✓ denotes using a pretrained model, while ✗ denotes training from scratch. Our proposed CrossWeaver family, trained from scratch, achieves a new state-of-the-art in both accuracy and efficiency.

Method	Pretrained	Modality	Backbone	car	truck	bus	van	freight	mAP ₅₀ (%)	mAP (%)	Params (M)	GFLOPs
YOLOv8s	✓	RGB	Single	91.9	68.8	92.7	56.5	53.1	72.6	56.8	11.2	28.6
YOLOv8s	✓	IR	Single	98.3	76.7	97.0	66.6	69.9	81.6	55.1	11.2	28.6
GM-DETR [36]	✓	RGB+IR	Shared	92.4	80.8	92.8	64.9	75.3	80.8	55.9	70.0	176.0
RGFNet [46]	✓	RGB+IR	Dual	98.7	80.2	95.5	67.9	76.6	83.8	-	79.8	-
DMM [49]	✓	RGB+IR	Dual	98.1	82.6	96.0	70.5	69.6	83.4	-	87.9	-
LoME [44]	✓	RGB+IR	Shared	98.5	84.3	96.5	71.6	79.3	86.0	64.7	45.5	83.7
ICAFusion [32]	✓	RGB+IR	Dual	98.0	84.2	97.1	67.1	73.8	84.8	63.1	120.2	-
COMO [13]	✓	RGB+IR	Dual	98.6	84.1	97.4	71.5	78.9	86.1	65.5	24.3	89.7
GAANet [48]	✗	RGB+IR	Dual	90.3	80.5	89.8	68.7	65.6	79.0	52.1	79.0	194.9
Fusion-Mamba [4]	✗	RGB+IR	Dual	96.7	80.2	95.3	64.0	59.8	79.2	56.0	287.6	-
MDPMFN [17]	✗	RGB+IR	Shared	98.0	80.9	95.5	67.7	64.3	81.3	-	-	41.5
UAVD-Mamba [11]	✗	RGB+IR	Dual	98.6	83.9	96.9	66.1	69.8	83.0	-	39.7	38.9
CrossWeaver-N (Ours)	✗	RGB+IR	Shared	98.6	80.5	96.7	69.5	76.8	84.4	62.7	5.41	19.5
CrossWeaver-T (Ours)	✗	RGB+IR	Shared	98.7	84.8	97.1	71.8	80.5	<u>86.5</u>	<u>65.0</u>	<u>9.44</u>	32.7
CrossWeaver-S (Ours)	✗	RGB+IR	Shared	98.7	84.4	97.4	72.9	80.6	86.8	65.5	19.67	65.7

Table 3. Ablation study on the rank selection strategy for the MCDA module. Our dynamic selection strategy outperforms all fixed-rank configurations, which utilizes ranks 9 to 6.

Rank Strategy	mAP ₅₀ (%)	mAP (%)	Params (M)	GFLOPs
$r = 9$	86.1	64.9	19.67	65.7
$r = 12$	86.5	65.0	19.82	65.7
$r = 6 \rightarrow 3$	85.4	64.3	19.52	65.7
$r = 12 \rightarrow 9$	86.3	64.7	19.82	65.7
$r = 9 \rightarrow 6$ (Ours)	86.8	65.5	19.67	65.7

our model family demonstrates exceptional scalability and efficiency. Our medium model CrossWeaver-T achieves a highly competitive 86.5% mAP₅₀, yet it uses only 9.44 million parameters, making it over 2.5 times more compact than COMO. Our smallest model CrossWeaver-N achieves an impressive 84.4% mAP₅₀ with only 5.41 million parameters and 19.5 GFLOPs, establishing a new benchmark for lightweight multimodal detectors.

4.5. Generality on Different Alignment Conditions

To validate the generality and robustness of our model, we conduct experiments on two benchmarks with varying alignment conditions in Table 4. On the well-aligned M3FD dataset, our CrossWeaver achieves a new state-of-the-art performance with a mAP₅₀ of 67.0%, establishing a significant 2.2% lead over the prior best method TFDet. Furthermore, our model demonstrates strong competitiveness in the highly demanding unaligned DVTOD dataset. CrossWeaver achieves 86.1% mAP₅₀ and only uses 19.7 million parameters. This combined success across both datasets directly validates our architecture’s ability to robustly balance modality similarity and complementarity while mitigating misalignment-induced quantization errors. This superior

performance relative to its parameter count highlights its significant efficiency advantage over competing methods.

4.6. Qualitative Results

The qualitative detection results for the DroneVehicle dataset are presented in Fig. 4. These results demonstrate our method’s performance across representative scenarios: daytime, well-lit night, and low-light night. The visualizations show that our method effectively detects targets. Notably, in the low-light scenario where vehicles are invisible in the RGB image, our method still detects them accurately. This highlights the effectiveness of the proposed method in multimodal object detection tasks, particularly in challenging remote sensing scenarios with variable illumination.

4.7. Deployment Analysis and Efficiency

We evaluate the real-world deployment performance on an NVIDIA Jetson NX. We develop the CrossWeaver-lite variant by replacing the deployment-hostile semantic fusion block based on Mamba with a simple, hardware-friendly add operation. The results in Table 5 validate this design. Our CrossWeaver-lite-S model achieves a highly competitive 85.6% mAP₅₀ while running at 46.30 FPS. Furthermore, our CrossWeaver-lite-N model shows outstanding scalability, reaching 56.39 FPS with only 3.50 million parameters. In stark contrast, other state-of-the-art methods like LoME, while achieving a high 86.0% mAP₅₀, are impractical for remote sensing as they run at only 13.22 FPS. This confirms our deployment-aware approach provides a superior accuracy-speed trade-off for edge applications.

4.8. Robustness to Positional Shift

To evaluate the model’s robustness against weak misalignment quantitatively, we conduct an experiment using simu-

Table 4. Comparison with State-of-the-Art methods. **Left:** M3FD dataset with well-aligned scenarios. **Right:** DVTOD dataset with unaligned scenarios. Our proposed CrossWeaver demonstrates superior performance and generality across both alignment conditions.

Model	person	car	bus	motor	lamp	truck	mAP ₅₀	mAP
EME [42]	68.4	81.2	63.4	43.9	35.8	53.5	57.7	37.0
CDDF [45]	-	-	-	-	-	-	61.9	38.6
DPAL [19]	71.5	83.3	69.0	44.8	44.5	57.4	61.7	-
TFDet [43]	-	-	-	-	-	-	64.8	41.0
CrossWeaver	82.7	87.5	60.4	59.2	55.1	71.0	67.0	42.1

Model	person	car	bicycle	mAP ₅₀	Params (M)
CMX [39]	88.9	75.9	79.6	81.6	132
CFT [28]	88.9	78.0	81.3	82.7	206
CMA-Det [33]	90.3	81.6	83.1	85.0	33.0
ICAFusion [32]	89.9	79.5	83.7	84.4	120
CrossWeaver	90.6	82.0	85.8	86.1	19.7



Figure 4. Visualization of detection results on the DroneVehicle dataset across three distinct scenarios: daytime, well-lit night, and dark night. The first two rows display the ground truth for the IR and RGB, while the bottom two rows show corresponding predictions, demonstrating robustness across all conditions.

Table 5. Deployment performance on the NVIDIA Jetson NX, benchmarked using TensorRT. Our CrossWeaver-lite models demonstrate a superior accuracy-speed trade-off for real-time edge inference.

Model	mAP ₅₀ (%)	Params (M)	GFLOPs	FPS (Hz)
ICAFusion [32]	84.8	80.55	120.2	9.13
LoME [44]	86.0	45.53	83.7	13.22
CrossWeaver-N (Ours)	83.4	3.50	6.5	56.39
CrossWeaver-T (Ours)	85.3	6.06	10.7	47.05
CrossWeaver-S (Ours)	85.6	12.34	20.2	46.30

lated positional offsets. The models are validated on shifted image pairs where RGB images are artificially shifted along the x-axis and y-axis within a range of 15 pixels, while

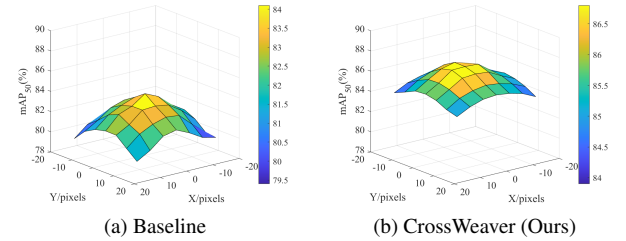


Figure 5. Surface plot visualization of the positional shift robustness. Horizontal coordinates represent the simulated pixel offsets, while the vertical coordinate shows the corresponding mAP₅₀.

the IR images remain stationary. The visualization in Figure 5 reveals the performance impact of these spatial shifts. The baseline model demonstrates clear sensitivity to misalignment, suffering a severe degradation as the offset approaches 15 pixels. In contrast, CrossWeaver on the right exhibits a much flatter and more stable performance surface. This result highlights the superior robustness of CrossWeaver under weakly aligned conditions.

5. Conclusion

In this work, we observe that mainstream multimodal detectors struggle to balance modality similarity and complementarity, and that weak alignment introduces quantization errors in multi-scale predictions. To address these issues, we propose CrossWeaver, a lightweight framework that reduces parameter redundancy and mitigates quantization errors while preserving robust detection performance. CrossWeaver adopts a shared-core and adaptive-refinement philosophy. Our work ultimately proves the feasibility of achieving superior detection accuracy alongside real-time processing capabilities on constrained edge platforms. This approach sets a new baseline for practical MMOD deployments. Future work will extend CrossWeaver to handle severe misalignments and unpaired data, aiming to significantly strengthen its robustness in real-world deployment.

Acknowledgment. This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant 62401049 and in part by China Postdoctoral Science Foundation Funded Project under Grant 2023M740268.

References

- [1] Chen Chen, Jiahao Qi, Xingyue Liu, Kangcheng Bin, Ruigang Fu, Xikun Hu, and Ping Zhong. Weakly misalignment-free adaptive feature alignment for uavs-based multimodal object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26836–26845, 2024. 2, 3
- [2] Yi-Ting Chen, Jinghao Shi, Zelin Ye, Christoph Mertz, Deva Ramanan, and Shu Kong. Multimodal object detection via probabilistic ensembling. In *European Conference on Computer Vision*, pages 139–158. Springer, 2022. 2
- [3] Jifeng Dai, Haozhi Qi, Yuwen Xiong, Yi Li, Guodong Zhang, Han Hu, and Yichen Wei. Deformable convolutional networks. In *Proceedings of the IEEE international conference on computer vision*, pages 764–773, 2017. 3, 4
- [4] Wenhao Dong, Haodong Zhu, Shaohui Lin, Xiaoyan Luo, Yunhang Shen, Guodong Guo, and Baochang Zhang. Fusion-mamba for cross-modality object detection. *IEEE Transactions on Multimedia*, 2025. 2, 7
- [5] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 1
- [6] Junjie Guo, Chenqiang Gao, Fangcen Liu, Deyu Meng, and Xinbo Gao. Damsdet: Dynamic adaptive multispectral detection transformer with competitive query selection and adaptive feature fusion. In *European Conference on Computer Vision*, pages 464–481. Springer, 2024. 1
- [7] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022. 4
- [8] Tao Huang, Xiaohuan Pei, Shan You, Fei Wang, Chen Qian, and Chang Xu. Localmamba: Visual state space model with windowed selective scan. In *European Conference on Computer Vision*, pages 12–22. Springer, 2024. 5
- [9] Soonmin Hwang, Jaesik Park, Namil Kim, Yukyung Choi, and In So Kweon. Multispectral pedestrian detection: Benchmark dataset and baseline. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1037–1045, 2015. 3
- [10] Glenn Jocher, Ayush Chaurasia, and Jing Qiu. YOLO by Ultralytics, 2023. 5
- [11] Wei Li, Jiaman Tang, Yang Li, Beihao Xia, Ligang Tan, and Hongmao Qin. Uavd-mamba: Deformable token fusion vision mamba for multimodal uav detection. *arXiv preprint arXiv:2507.00849*, 2025. 3, 7
- [12] Siyuan Liang, Hao Wu, Li Zhen, Qiaozhi Hua, Sahil Garg, Georges Kaddoum, Mohammad Mehdi Hassan, and Keping Yu. Edge yolo: Real-time intelligent object detection system based on edge-cloud cooperation in autonomous vehicles. *IEEE Transactions on Intelligent Transportation Systems*, 23(12):25345–25360, 2022. 1
- [13] Chang Liu, Xin Ma, Xiaochen Yang, Yuxiang Zhang, and Yanni Dong. Como: Cross-mamba interaction and offset-guided fusion for multimodal object detection. *arXiv preprint arXiv:2405.08012*, 2024. 3, 6, 7
- [14] Chang Liu, Xin Ma, Xiaochen Yang, Yuxiang Zhang, and Yanni Dong. Como: Cross-mamba interaction and offset-guided fusion for multimodal object detection. *Information Fusion*, page 103414, 2025. 1
- [15] Jingjing Liu, Shaoting Zhang, Shu Wang, and Dimitris N Metaxas. Multispectral deep neural networks for pedestrian detection. In *British Machine Vision Conference*, 2016. 3
- [16] Jinyuan Liu, Xin Fan, Zhanbo Huang, Guanyao Wu, Risheng Liu, Wei Zhong, and Zhongxuan Luo. Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5802–5811, 2022. 5
- [17] Kewei Liu, Dongliang Peng, and Tao Li. Multimodal remote sensing object detection based on prior-enhanced mixture-of-experts fusion network. *IEEE Transactions on Geoscience and Remote Sensing*, 2025. 7
- [18] Yue Liu, Yunjie Tian, Yuzhong Zhao, Hongtian Yu, Lingxi Xie, Yaowei Wang, Qixiang Ye, Jianbin Jiao, and Yunfan Liu. Vmamba: Visual state space model. *Advances in neural information processing systems*, 37:103031–103063, 2024. 5
- [19] Yanfeng Liu, Wei Guo, Chaojun Yao, and Lefei Zhang. Dual-perspective alignment learning for multimodal remote sensing object detection. *IEEE Transactions on Geoscience and Remote Sensing*, 2025. 8
- [20] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin Transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10012–10022, 2021. 1
- [21] Qi Ming and Xiaowu Xiao. Towards accurate medical image segmentation with gradient-optimized dice loss. *IEEE Signal Processing Letters*, 31:191–195, 2024. 1
- [22] Qi Ming, Lingjuan Miao, Zhiqiang Zhou, Junjie Song, and Aleksandra Pizurica. Gradient calibration loss for fast and accurate oriented bounding box regression. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–15, 2024. 1
- [23] Qi Ming, Lingjuan Miao, Zhiqiang Zhou, Nicolas Vercheval, and Aleksandra Pizurica. Not all boxes are equal: Learning to optimize bounding boxes with discriminative distributions in optical remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–14, 2024. 1
- [24] Sayan Nag. Image registration techniques: A survey. *arXiv preprint arXiv:1712.07540*, 2017. 3
- [25] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018. 3
- [26] Chunxia Qin, Zhenrong Zhang, Pengfei Hu, Chenyu Liu, Jiefeng Ma, and Jun Du. Svm3: A fast and robust approach to table separation line detection. *arXiv preprint arXiv:2405.11862*, 2024. 1
- [27] Chunxia Qin, Chenyu Liu, Pengcheng Xia, Jun Du, Baocai Yin, Bing Yin, and Cong Liu. Tdatr: Improving end-to-end table recognition via table detail-aware learning and cell-level visual alignment. *arXiv preprint arXiv:2603.22819*, 2026. 1

- [28] Fang Qingyun, Hu Dapeng, and Wang Zhaokui. Cross-modality fusion transformer for multispectral object detection. *arXiv preprint arXiv:2111.00273*, 2021. 3, 8
- [29] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 779–788, 2016. 1, 3
- [30] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(6):1137–1149, 2016. 1, 3
- [31] Marc-Michel Rohé, Manasi Datar, Tobias Heimann, Maxime Sermesant, and Xavier Pennec. Svf-net: Learning deformable image registration using shape matching. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 266–274. Springer, 2017. 3
- [32] Jiaming Shen, Yuyu Chen, Yuyang Liu, Xianjing Zuo, Huan Fan, and Wenkai Yang. Icafusion: Iterative cross-attention guided feature fusion for multispectral object detection. *Pattern Recognition*, 145:109913, 2024. 2, 3, 7, 8
- [33] Kechen Song, Xiaotong Xue, Hongwei Wen, Yingying Ji, Yunhui Yan, and Qinggang Meng. Misaligned visible-thermal object detection: A drone-based benchmark and baseline. *IEEE Transactions on Intelligent Vehicles*, 2024. 6, 8
- [34] Yiming Sun, Bing Cao, Pengfei Zhu, and Qinghua Hu. Drone-based rgb-infrared cross-modality vehicle detection via uncertainty-aware learning. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(10):6700–6713, 2022. 2, 5
- [35] Yiming Xiao, Fanman Meng, Qingbo Wu, Linfeng Xu, Mingzhou He, and Hongliang Li. Gm-detr: Generalized multispectral detection transformer with efficient fusion encoder for visible-infrared detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5541–5549, 2024. 2
- [36] Yuhang Xiao, Fanman Meng, Qingmin Wu, Linfeng Xu, Minlong He, and Hongliang Li. Gm-detr: Generalized multispectral detection transformer with efficient fusion encoder for visible-infrared detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 5541–5549, 2024. 7
- [37] Maoxun Yuan, Yinyan Wang, and Xingxing Wei. Translation, scale and rotation: Cross-modal alignment meets rgb-infrared vehicle detection. In *European Conference on Computer Vision*, pages 509–525. Springer, 2022. 1
- [38] Haote Zhang, Lipeng Gu, Wuzhou Quan, Fu Lee Wang, Honghui Fan, Jiali Tang, Dingkun Zhu, Haoran Xie, Xiaoping Zhang, and Mingqiang Wei. Wavelet-guided misalignment-aware network for visible-infrared object detection. *arXiv preprint arXiv:2507.20146*, 2025. 3
- [39] Jiaming Zhang, Huayao Liu, Kailun Yang, Xinxin Hu, Ruiping Liu, and Rainer Stiefelhagen. Cmx: Cross-modal fusion for rgb-x semantic segmentation with transformers. *IEEE Transactions on intelligent transportation systems*, 24(12):14679–14694, 2023. 8
- [40] Jiaqing Zhang, Mingxiang Cao, Weiyang Xie, Jie Lei, Daixun Li, Wenbo Huang, Yunsong Li, and Xue Yang. E2emfd: Towards end-to-end synchronous multimodal fusion detection. *Advances in Neural Information Processing Systems*, 37:52296–52322, 2024. 1
- [41] Lu Zhang, Xiangyu Zhu, Xiangyu Chen, Xu Yang, Zhen Lei, and Zhiyong Liu. Weakly aligned cross-modal learning for multispectral pedestrian detection. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 5127–5137, 2019. 3
- [42] Xue Zhang, Si-Yuan Cao, Fang Wang, Runmin Zhang, Zhe Wu, Xiaohan Zhang, Xiaokai Bai, and Hui-Liang Shen. Rethinking early-fusion strategies for improved multispectral object detection. *IEEE Transactions on Intelligent Vehicles*, 2024. 6, 8
- [43] Xue Zhang, Xiaohan Zhang, Jiangtao Wang, Jiacheng Ying, Zehua Sheng, Heng Yu, Chunguang Li, and Hui-Liang Shen. Tfdet: Target-aware fusion for rgb-t pedestrian detection. *IEEE Transactions on Neural Networks and Learning Systems*, 2024. 8
- [44] Yusi Zhang, Weiyang Xie, Tianlin Hui, Daixun Li, Jiaqing Zhang, Jie Lei, Yunsong Li, and Leyuan Fang. Lome: Lora-driven multimodal extractor for rgb-x vision tasks. *IEEE Transactions on Circuits and Systems for Video Technology*, 2025. 2, 7, 8
- [45] Zixiang Zhao, Haowen Bai, Jiangshe Zhang, Yulun Zhang, Shuang Xu, Zudi Lin, Radu Timofte, and Luc Van Gool. Cddfuse: Correlation-driven dual-branch feature decomposition for multi-modality image fusion. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5906–5916, 2023. 8
- [46] Zhicheng Zhao, Wei Zhang, Yun Xiao, Chenglong Li, and Jin Tang. Reflectance-guided progressive feature alignment network for all-day uav object detection. *IEEE Transactions on Geoscience and Remote Sensing*, 2025. 7
- [47] Yu Zheng, Danyang Zhang, Sinan Xie, Jiwen Lu, and Jie Zhou. Rotation-robust intersection over union for 3D object detection. In *European Conference on Computer Vision*, pages 464–480. Springer, 2020. 1
- [48] Zihao Zheng, Mingliang Zhou, Zhaowei Shang, Xuekai Wei, Huayan Pu, Jun Luo, and Weijia Jia. Gaanet: Graph aggregation alignment feature fusion for multispectral object detection. *IEEE Transactions on Industrial Informatics*, 2025. 7
- [49] Minghang Zhou, Tianyu Li, Chaofan Qiao, Dongyu Xie, Guoqing Wang, Ningjuan Ruan, Lin Mei, Yang Yang, and Heng Tao Shen. Dmm: Disparity-guided multispectral mamba for oriented object detection in remote sensing. *IEEE Transactions on Geoscience and Remote Sensing*, 63:1–13, 2025. 3, 7
- [50] Barbara Zitová and Jan Flusser. Image registration methods: a survey. *Image and vision computing*, 21(11):977–1000, 2003. 3